



KNEES AND THIGHS

The skeletal structure of the leg and knee is made up of the tibia and fibula (lower leg), the femur (upper leg), and the patella (knee cap) between the two limbs. These long bones in the lower and upper leg form the major lever system that allows the body to use the muscles of this region in all locomotive movements.

The knee joint is the only major joint between the bones of the lower and upper leg. It is classified as a hinge joint, and it allows only two major movements: flexion and extension. The range of motion, or the degree of freedom to move this joint, depends on both the bone structure and the flexibility of the muscle tissue, tendons, and ligaments that surround this joint. Typically, the knee joint is rather limited in movement compared with other joints in the body, but the combination of the knee and the hip joint allows us to perform a variety of complicated movements and can enhance various sports and leisure activities. The more flexible these muscles are, the more freedom of movement possible.

The knee is surrounded by several ligaments and tendons (figure 3.1) that bring more stability. In spite of these additional supportive structures, the knee is still quite vulnerable to injury. One of the most important ligaments around the knee is the patellar ligament. It extends from the patella to the upper front tibia. The tendons of the quadriceps muscles, located in the front of the thigh, blend with the patellar ligament, which attaches these muscles to the tibia. The medial collateral ligament supports the medial (inner) side of the knee, and the lateral (outer) side of the knee is supported by the lateral collateral ligament. The anterior and posterior cruciate ligaments help prevent anterior and posterior displacements of the femur on the tibia bone. These ligaments are located inside the knee and hold the tibia and femur bones together. The oblique popliteal and arcuate popliteal ligaments provide additional support to the lateral posterior (outer back) area of the knee.

In addition, the medial and lateral patellar retinacula also arise from the quadriceps tendon and contribute to anterior support of the knee. Finally, a meniscus sits on the plateau (top) of the tibia, which gives additional stability

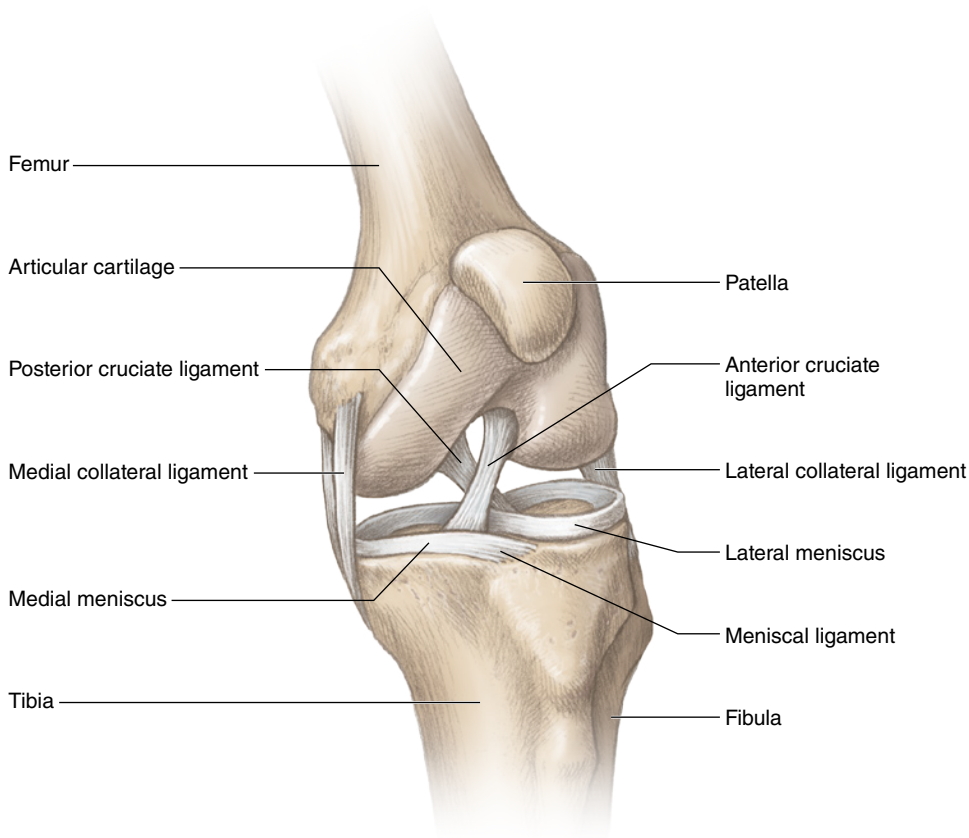


FIGURE 3.1 Knee ligaments and tissue.

to the knee and cushions the bones during walking, running, and jumping. Wear and tear of these menisci bring pain most often to the medial (inner) side of the knee joint.

Most of the muscles that control the movements of the knee are found in the thigh. However, a few calf muscles are also involved. Generally, the thigh muscles that move the knee are categorized into two groups. The four large anterior thigh muscles—rectus femoris, vastus intermedius, vastus lateralis, and vastus medialis—are collectively called the quadriceps muscles, and these are the major knee extensors. The large posterior thigh muscles—biceps femoris, semimembranosus, and semitendinosus—are collectively called the hamstring muscles, and these are the major knee flexors. The hamstrings are assisted in knee flexion by the gracilis and sartorius on the medial side of the thigh and the gastrocnemius, popliteus, and plantaris on the posterior side of the lower leg.

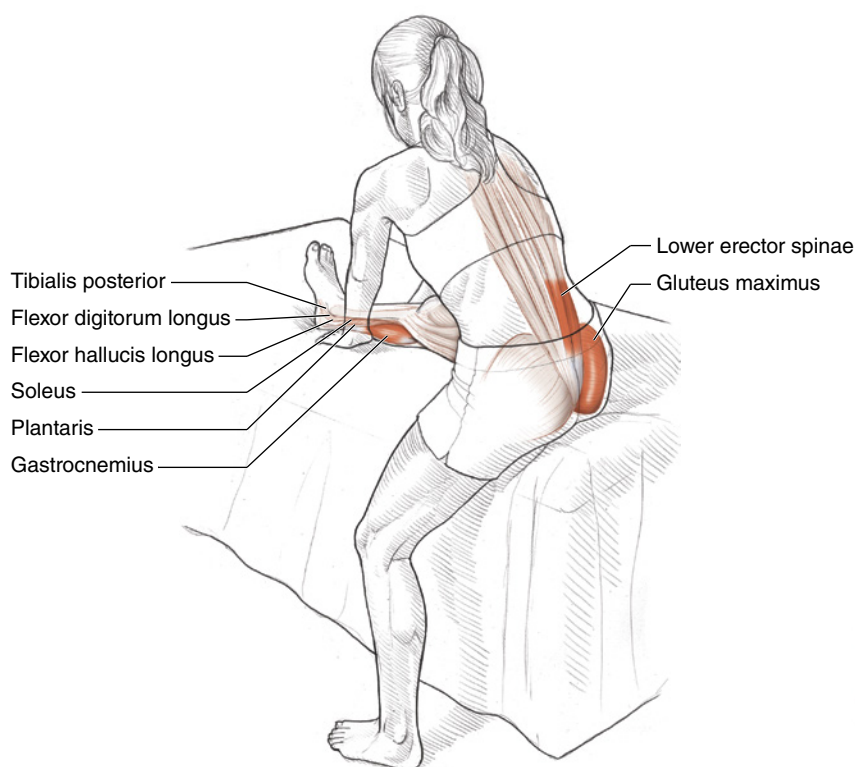
Flexion and extension are the two major movements of the knee. Most muscles in the body cross several joints, and thus many of these muscles are

able to perform several movements. Three of the quadriceps muscles, the vastus muscles, cross only one joint. This muscular arrangement allows these muscles to perform only knee extension. These three vastus muscles are strong extensors and sometimes may be sore and tight in front of the knee where the patella bone is located. Muscle tightness caused by lack of stretching of the quadriceps muscles is most often the cause of this problem. The knee extensors tend to exert less movement in walking, running, and jumping than the hamstring muscles do. On the other hand, the hamstring muscles create two major movements—knee flexion and hip extension—and are active during any locomotive movement of the body. Thus, it appears that more total load is put on the hamstring muscles than on the quadriceps muscles. Because of this factor, the hamstring muscles tend to become more fatigued and sore than the quadriceps muscles do during daily activities.

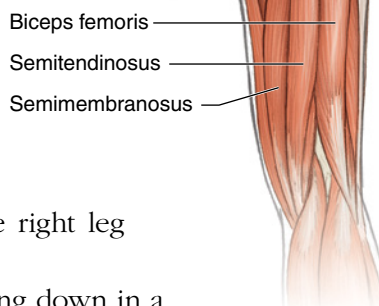
The muscles of the thigh that control the knee are important in all motor movements. Being much larger than the muscles of the calf and foot, the thigh muscles are better able to withstand muscular stress. Hence, muscular soreness occurs less often in these muscle groups. It is important, however, to have the right balance of strength and flexibility between the opposing muscle groups of the thigh. Most people have stronger but less flexible quadriceps muscles than hamstring muscles. People tend to stretch the hamstring muscles much more than the quadriceps muscles. This creates an imbalance between the two muscle groups. Chronic overstretching of the hamstrings without comparable stretching of the quadriceps can cause more harm than good. This is the reason hamstring muscles are sore more often than quadriceps muscles are. Overstretching can also lead to chronic fatigue and a decrease in strength in the hamstring muscles. To correct this imbalance, you need to put more emphasis on quadriceps stretching and decrease the emphasis on hamstring stretching.

People often sit in one position for a long time, especially when in a car, sitting at a desk, or on an airplane. Thus, it is not surprising that after sitting for hours, people feel the need to get up and stretch. When people do stand after long periods of sitting, they typically find that their joints and muscles have become temporarily stiff. Most often they feel more stiffness in the knee joint, and getting up after sitting a long time can be a rather painful experience. Because of this, it is recommended to get up often during those long sitting hours and move around. Stretching these muscles is a natural remedy. Many people have found that stretching and moving the leg muscles provide relief from muscular and joint tension and pain. Because muscular soreness and tension are common in the thigh muscles, both temporary and lasting relief can be obtained from a daily stretching routine that is a consistent part of a fitness program.

BEGINNER SEATED KNEE FLEXOR STRETCH



Hamstrings:



Execution

1. Sit on a couch, bed, or bench, with the right leg extended on the surface.
2. Rest the left foot on the floor, or let it hang down in a relaxed manner.
3. Place the hands on the couch, bed, or bench next to the right thigh or knee.
4. Bend at the waist and lower the head toward the right knee, keeping the back of the right knee comfortably on the couch, bed, or bench as much as possible.
5. While bending forward, slide the hands toward the right foot, keeping them alongside the lower leg.
6. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right hamstring (semitendinosus, semimembranosus, biceps femoris), right gluteus maximus, right gastrocnemius, right lower erector spinae (iliocostalis, spinalis, longissimus)

Less-stretched muscles: Right soleus, right plantaris, right popliteus, right flexor digitorum longus, right flexor hallucis longus, right tibialis posterior

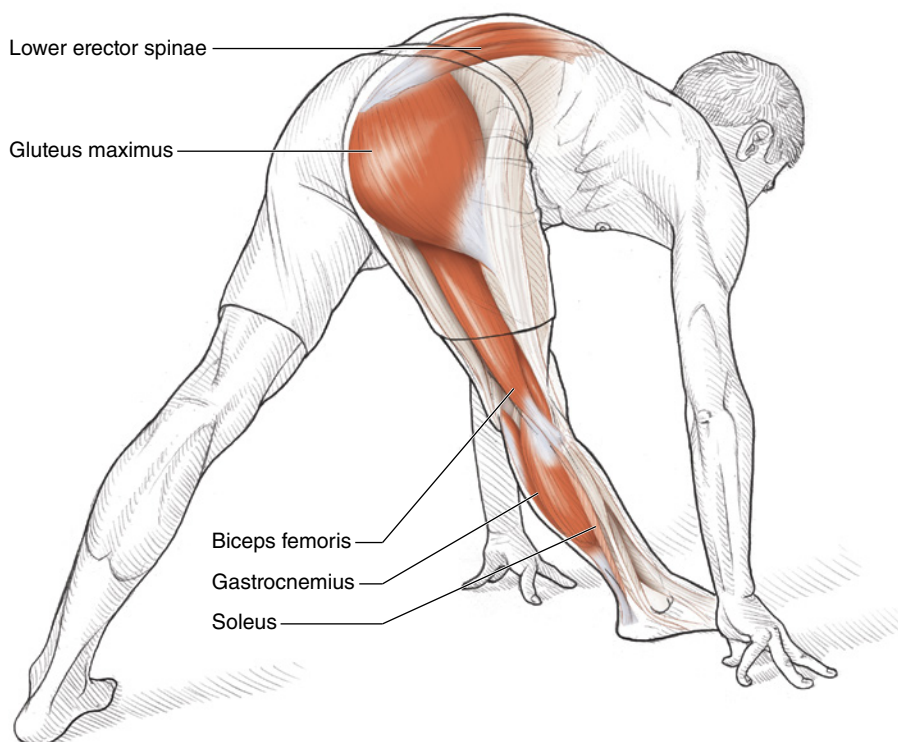
Stretch Notes

Tight knee flexors, or hamstring muscles, affect posture and the way the body moves during exercise. When these muscles are tight, the pelvis and hips are pulled out of their natural alignment, resulting in a flattened back and loss of the natural curve. A flatter lower back puts increased pressure on the sciatic nerve that runs down the legs and can cause muscles to tighten more. When the muscles are tight, they are also short, and short knee flexors increase the strain on the lower-trunk extensor muscles, especially when you bend forward at the waist. This added strain then injures the lower-trunk extensor muscles and is one of the most common causes of a sore lower back. Also, a lack of flexibility in the knee flexors makes these muscles more injury prone when a person suddenly increases movement speed or experiences greater workloads.

There are many reasons an inactive person might have short knee flexor muscles. First, a person could be born with naturally short hamstrings. Second, the hamstrings can become short if a person sits for long periods. Regardless of the reason, hamstrings can become longer after performing regular stretching exercises.

Stretching the knee flexors one leg at a time reduces the stress on the legs and back. The knee flexor stretch can be performed on a soft couch or other soft surface and can be done at any time—while sitting on the couch watching TV or just relaxing after a long day's work. Doing this stretching exercise from a seated position, with one leg up on the couch surface and the other leg hanging down allows you to concentrate solely on stretching these muscles and allowing the other muscles in the body to relax. If you are not flexible or are beginning a stretching program, it would perhaps be better to start this stretch with your right knee slightly bent and then gradually work on straightening the knee as your flexibility improves. If you want to maximize the stretch of these muscles, start working with a straight knee position. While executing this stretch, try to avoid tilting the pelvis forward or curving the back. It is also beneficial to bend the trunk forward as a single unit, keeping it centered above or next to the side of the right thigh.

INTERMEDIATE STANDING KNEE FLEXOR STRETCH



Execution

1. Stand upright with the right heel a comfortable distance ahead of the left toes.
2. Keeping the right knee straight and the left knee slightly bent, bend the trunk over toward the right knee.
3. Reach the hands toward the right foot.
4. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right hamstring (semitendinosus, semimembranosus, biceps femoris), right gluteus maximus, right gastrocnemius, lower right erector spinae (iliocostalis, spinalis, longissimus)

Less-stretched muscles: Right soleus, right plantaris, right popliteus, right flexor digitorum longus, right flexor hallucis longus, right tibialis posterior

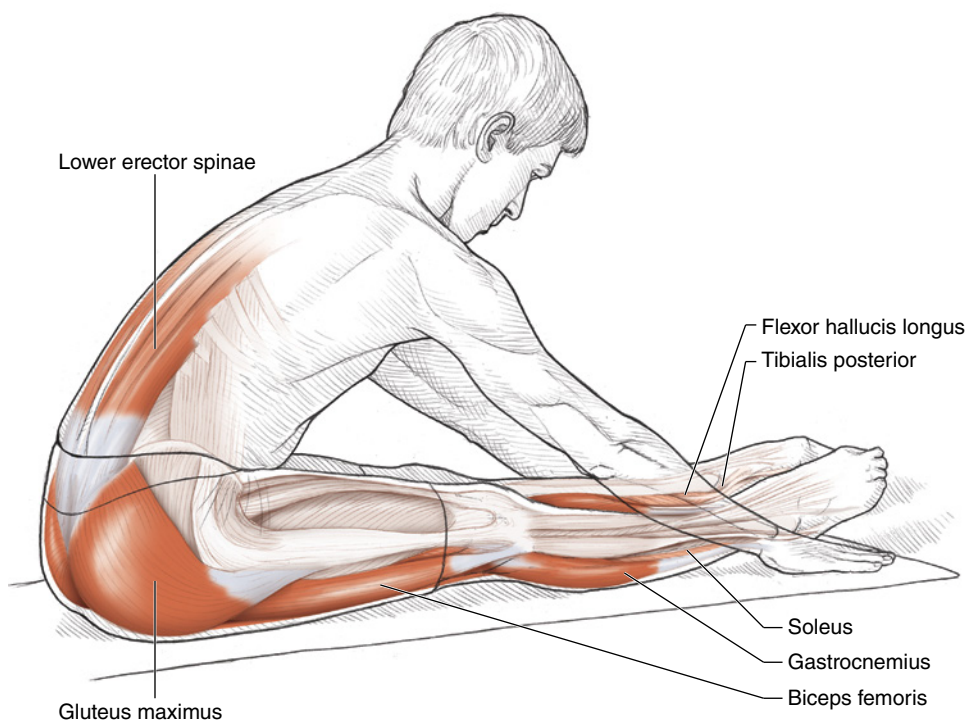
Stretch Notes

If you start participating in a sport and do not stretch properly, it is more likely your hamstrings will tighten up. Tight hamstrings are common among both distance runners and sprinters who have significantly increased their speed, the distance run, or the amount of uphill climb. Tightness in the muscles can ease away during exercise as the muscles get warmer, but after stopping it can return. Also, tightness is often an indicator of minor or major muscle strains, a common occurrence mainly felt after exercise. In addition, muscle strength imbalances, in which the knee extensors are stronger or the gluteal muscles are weaker than the hamstrings, will also cause tightness. Thus, it is especially important to stretch properly after exercise because this is when the muscles are warm and more receptive to stretching.

This is the most commonly used stretch for the hamstring muscles. It can easily be done at any time you feel the need to stretch your hamstrings. After any type of fitness activity, minor aches and tightness in the hamstring muscles are possible. It is not unusual to experience such discomfort after almost any exercise session. This is the optimal time to lightly stretch these muscles. In most cases, this stretch will relieve those uncomfortable symptoms, and you will be able to continue on to your other daily routines without concern about your muscle condition.

For the best results in this stretch, try to keep the right knee straight and bend the torso directly from the hip. It is also important to keep the back as straight as possible when executing this stretch. Turning the right foot out slightly and bending the head and trunk more toward the medial (inner) side of the right knee will increase the stretch of the biceps femoris, which is located on the back outer side of the thigh. On the other hand, turning the right foot in slightly and bending the head and trunk more toward the lateral (outer) side of the knee will increase the stretch of the semitendinosus and semimembranosus muscles, located on the back inner side of the thigh.

ADVANCED SEATED KNEE FLEXOR STRETCH



Execution

1. Sit on the floor, rug, or exercise mat with both legs extended and the insides of the ankles as close together as possible.
2. Keep the feet relaxed in a natural position.
3. Place the hands on the floor next to the thighs.
4. Bend at the waist and lower the head toward the legs. If possible, keep the back of the knees on the floor.
5. While bending forward, slide the hands toward the feet, and keep them alongside the legs.

Muscles Stretched

Most-stretched muscles: Hamstrings (semitendinosus, semimembranosus, biceps femoris), gluteus maximus, gastrocnemius, lower erector spinae (iliocostalis, spinalis, longissimus)

Less-stretched muscles: Soleus, plantaris, popliteus, flexor digitorum longus, flexor hallucis longus, tibialis posterior

Stretch Notes

When the hamstrings are tight, the pelvis and hips are pulled out of their natural alignment, resulting in a flattened back and loss of the natural curve. A flatter lower back puts increased pressure on the sciatic nerve that runs down the legs and can cause muscles to tighten more. Tight muscles are also short muscles, and short knee flexors increase the strain on the lower-trunk extensor muscles, especially when you bend forward at the waist. This added strain then injures the lower-trunk extensor muscles, one of the most common causes of a sore lower back. A tight muscle can compress the blood vessels within the muscle, and the reduced blood flow can make the hamstrings and lower-back extensors tighter and more fatigued.

Although this stretch helps alleviate problems by increasing flexibility, it is not recommended until you have already increased your hamstring flexibility. If this exercise is done when both sets of muscles are tight, you risk causing damage to the lower back. This is because the hamstrings are usually larger and stronger, and so the weaker link gives out first.

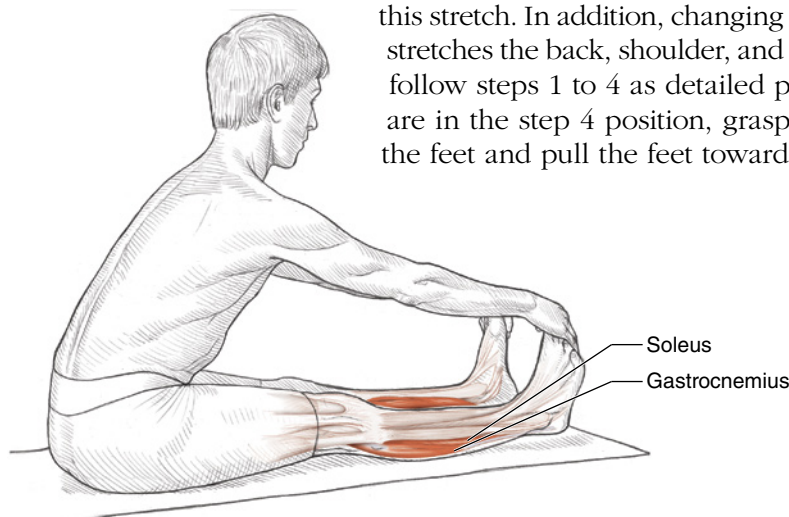
In this exercise, you are able to stretch both legs at the same time. For you to stretch better and improve your flexibility, try your best to keep the knees straight. It is also important to keep the back straight. When you bend the trunk forward, try to move it as a single unit, keeping it centered between your legs. Following these procedures allows you to stretch the hamstring muscles more effectively and brings you more enjoyable, quicker, and better results. It is usually more comfortable to perform this stretch on a carpet, exercise mat, or other soft surface. Doing this stretching exercise in a seated position allows you to relax the other muscles in your body. This stretch could easily be performed while you are just sitting around watching TV or reading or doing any seated leisure activity. Because we all do a lot of sitting in a day, this stretch can be done at any time and repeated throughout the day. A concentrated effort to remind yourself to perform this stretch daily might be helpful in accomplishing this goal.

VARIATION

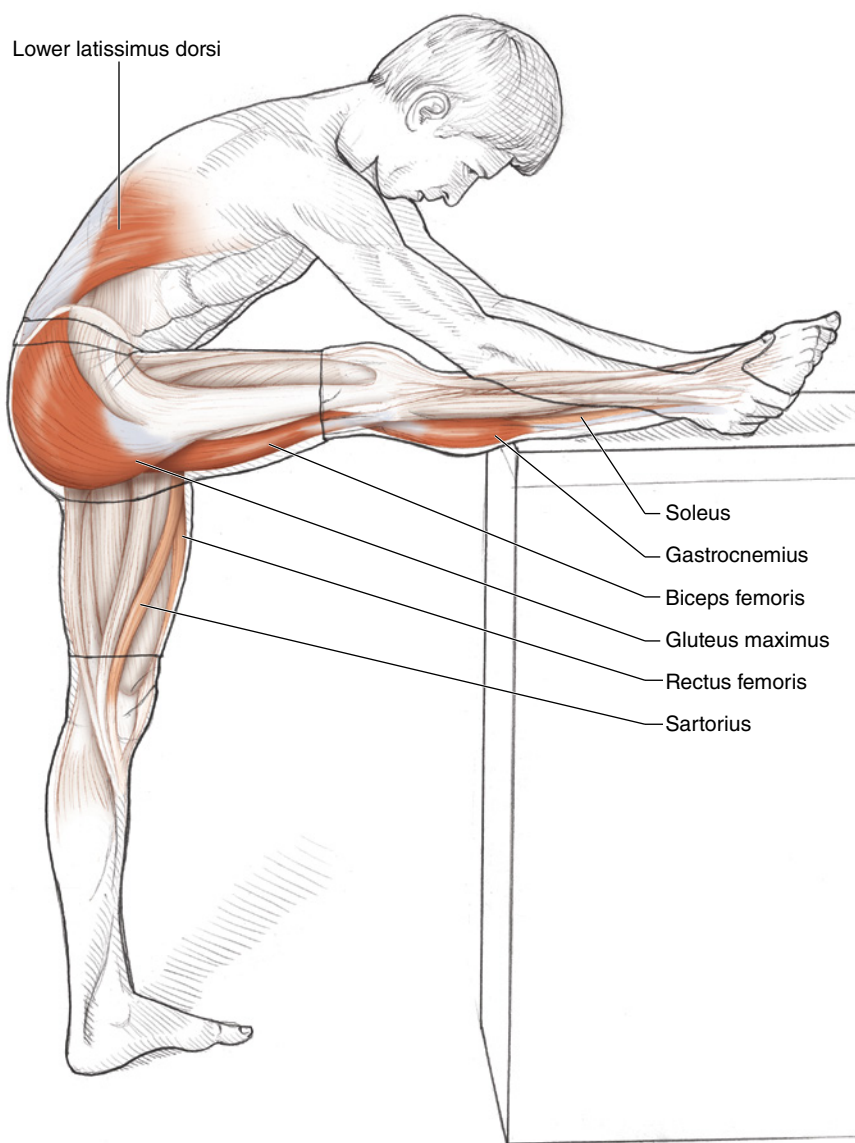
Seated Knee, Ankle, Shoulder, and Back Stretch

Instead of leaving the hands alongside the legs, if you grasp the toes and pull them slowly toward the knees (dorsiflexed position), you add the calf muscles to

this stretch. In addition, changing to this hand position stretches the back, shoulder, and arm muscles. Simply follow steps 1 to 4 as detailed previously. Once you are in the step 4 position, grasp the toes or balls of the feet and pull the feet toward the knees.



EXPERT RAISED-LEG KNEE FLEXOR STRETCH



Execution

1. Stand upright with your weight balanced on the left leg.
2. Flex the right hip and place the right leg, with the knee straight, on a table, bench, or other stable object that is higher than the height of the hips.
3. Bend at the waist, extend your arms over the lower right leg, and lower the head toward the right leg, keeping the right knee as straight as possible.
4. Keep the left knee straight and the left foot pointing in the same direction as the right leg.
5. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right gluteus maximus, right hamstring (semitendinosus, semimembranosus, biceps femoris), right erector spinae (iliocostalis, spinalis, longissimus), lower right latissimus dorsi, right gastrocnemius

Less-stretched muscles: Right soleus, right popliteus, right plantaris, right flexor digitorum longus, right flexor hallucis longus, right tibialis posterior, left sartorius, left rectus femoris

Stretch Notes

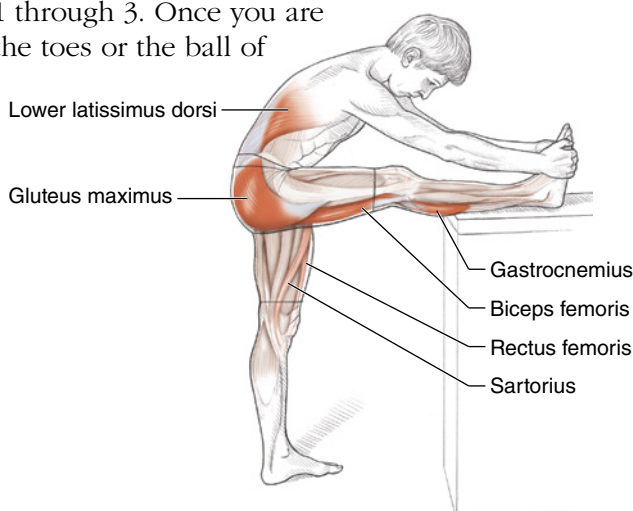
This is a more advanced stretch for those whose knee flexors are already more flexible than those of the average athlete. Be sure to select the right starting height for the table, bench, couch, or other stable object you will put your leg on. At the beginning of your stretching program, it is recommended to start at a lower height based on your state of flexibility and then periodically increase the height of the surface by several inches as your flexibility improves. Increasing the height of the surface by 1 to 2 feet (30 to 60 cm) above the hips as your flexibility improves will increase the stretch of these muscle groups. At this point, you will also start feeling a stretch to a portion of the front part of the left-leg muscle groups (sartorius; rectus femoris; vastus intermedius, lateralis, and medialis) as you increase the table to the highest possible height.

To maximize the stretch of the knee flexors, do not bend the knees, tilt the pelvis forward, or curve the back. In addition, bend the trunk straight forward as a single unit, keeping it centered over the right leg.

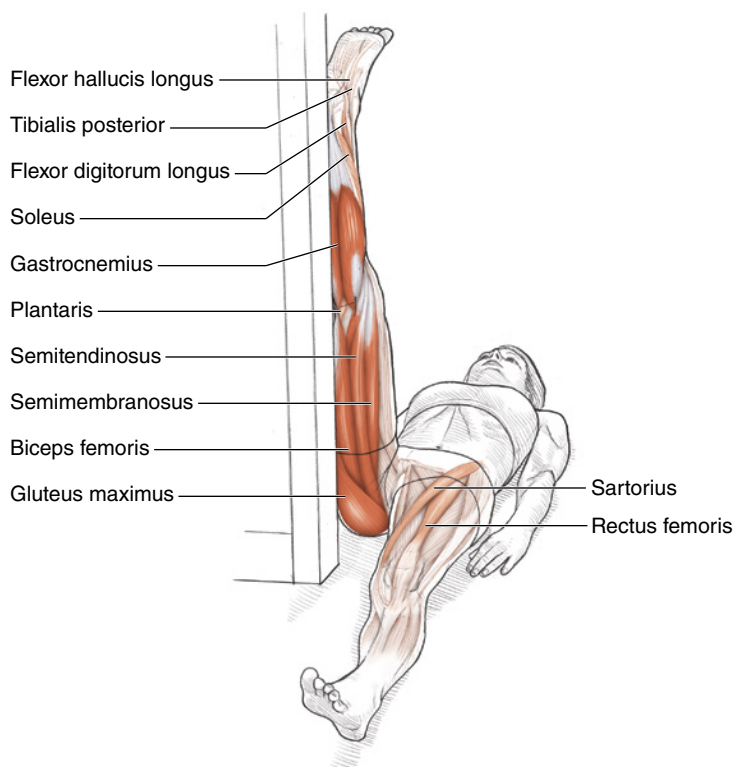
VARIATION

Raised-Leg Knee, Ankle, Shoulder, and Back Stretch

Grasping and pulling on the toes adds more muscles to the stretching process. This combination exercise stretches most of the posterior (back) body muscles at the same time. This naturally saves time if your exercise time is limited. Follow steps 1 through 3. Once you are in the step 4 position, grasp the toes or the ball of the foot and pull the foot toward the knee.



RECUMBENT KNEE FLEXOR STRETCH



Execution

1. Lie flat on your back in a doorway, with the hips placed in front of the doorframe.
2. Raise the right leg and rest it on the doorframe. Keep the right knee straight and the left leg flat on the floor.
3. Place the hands palms down on either side of the buttocks.
4. Keeping the right leg straight, use the hands to slowly move the buttocks through the doorframe until you feel a stretch in the back of the leg.
5. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right gluteus maximus, right hamstring (semitendinosus, semimembranosus, biceps femoris), right gastrocnemius

Less-stretched muscles: Right soleus, right popliteus, right plantaris, right flexor digitorum longus, right flexor hallucis longus, right tibialis posterior, left sartorius, left rectus femoris

Stretch Notes

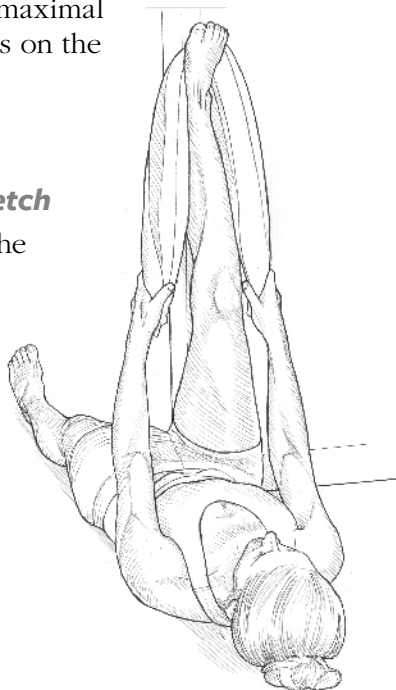
When stretching the knee flexors, you must pay attention to the lower back. If the lower-back extensor muscles are tight, they will limit the ability to perform most knee flexor stretches. As a result, many people overstress the back. Also it is easy to tilt the pelvis forward or curve the back. Doing this can further harm the lower-back muscles. When you are lying on your back, it is easier to maintain correct back positioning, and the floor provides additional back support. Thus, this exercise is the best knee flexor stretch to use when you have back problems.

Positioning your body in the right place for this stretch could take extra time and effort, but once you are able to find the right position, it is an excellent stretch. To maximize the stretch of the knee flexors, do not bend the knees, tilt the pelvis forward, or round the back. Adjust the distance between the buttocks and the doorframe to increase or decrease the stretch. The closer the buttocks are to the doorframe, the greater the stretch. Once the buttocks cannot be positioned any closer to the doorframe, bending the right leg at the hip and moving the right leg toward the head can increase the stretch. It is also important to keep the left leg straight in front of you on the floor in order for you to get the maximal effect of this stretch. When reaching the maximal stretch limit, you will find that the quadriceps muscles on the left leg are being stretched as well.

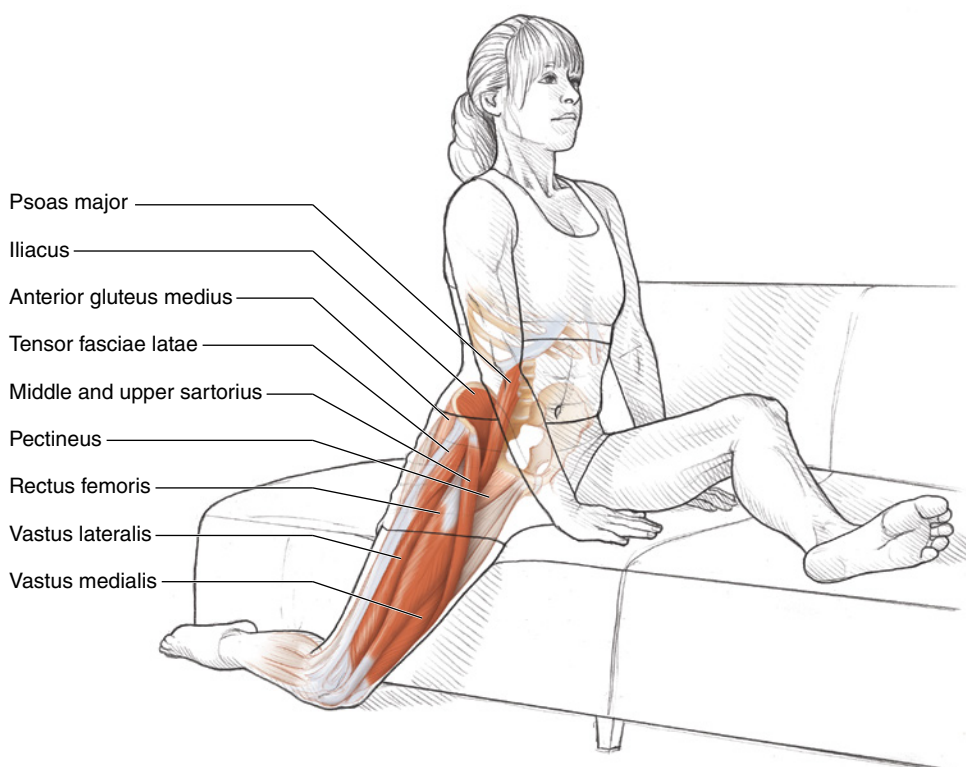
VARIATION

Recumbent Knee, Ankle, Shoulder, and Back Stretch

Using a towel to pull the toes downward increases the number of muscles being stretched. Follow steps 1 to 4. Once you feel the hamstrings being stretched, use the towel to pull the toes and foot toward the floor. This combo exercise stretches most of the muscles on the back side of the body at the same time. In other words, you can work on multiple muscle groups, including the calf, hamstring, back, shoulder, and arms, to a small degree. This naturally saves time if your exercise time is limited.



BEGINNER SEATED KNEE EXTENSOR STRETCH



Execution

1. Sit upright on a couch or bed, with the left knee bent at less than a 90-degree angle in front of you. The lateral side of the left leg should be flat on the surface and the left hip near the edge of the couch or bed.
2. Balance the weight of your body over the left hip.
3. Extend the right leg behind the torso, and touch the floor with the right knee. The lower right leg lies on the floor.
4. Place the hands on the couch or bed to maintain balance.
5. Move the hips slowly forward, if needed, for more stretch.
6. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right quadriceps (rectus femoris, vastus medialis, vastus intermedius, vastus lateralis), middle and upper right sartorius, right psoas major, right iliacus, right tensor fasciae latae

Less-stretched muscles: Right pectineus, anterior right gluteus medius

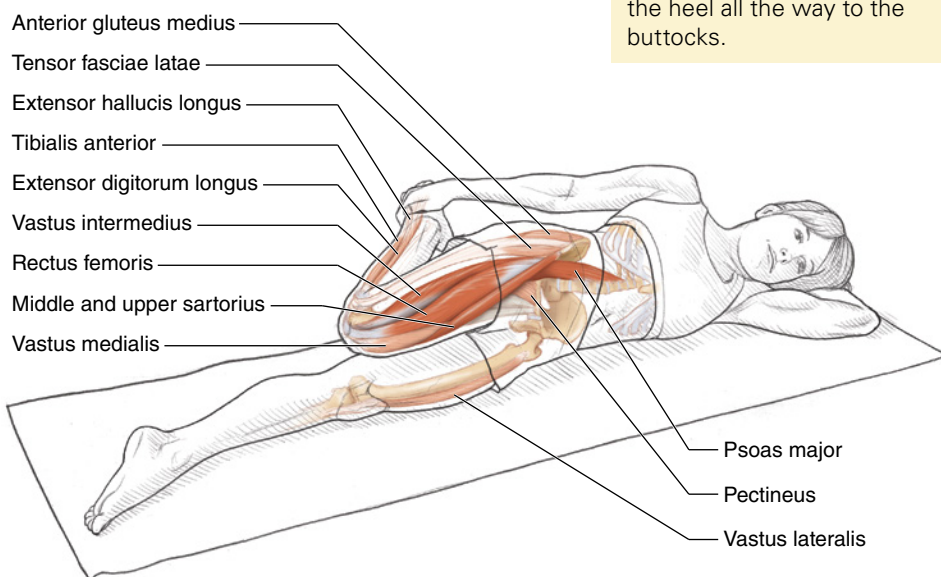
Stretch Notes

The knee extensors, the quadriceps, are used for common actions such as standing, sitting, walking, running, and jumping. Strains and injuries of the quadriceps muscles and tendons are common among athletes 15 to 30 years old who are engaged in explosive activities. On the other hand, for people engaged in daily living activities, the average age for injuries to these muscles is 65. Muscle strains and tears usually happen when a muscle is stretched beyond its limit, tearing the muscle fibers. They frequently occur near the point where the muscle joins the tendon. The four main causes of quadriceps injuries are muscle tightness, muscle imbalance, poor conditioning, and muscle fatigue. The ease of performing this beginning-level stretch will hopefully motivate you to stretch these muscles, especially because it can be done while reading, watching TV, or just relaxing.

This is a beginner's stretch for the quadriceps muscles. You can perform this stretch while sitting on the edge of a sofa or bed. The sitting position helps make the execution of this stretch more comfortable and relaxing. Place a pillow under the right knee for added comfort. Having the left leg in the bent position in front of you allows the stretch to focus on the right leg's quadriceps muscles. Extend the right leg back from the torso.

Moving your hips slowly forward allows you to monitor the amount of stretch you put on the quadriceps muscles. This stretch can increase in intensity as needed or desired. Simply try arching the back slightly while moving the hips in the forward direction. After maximizing this level of stretching, start using the more advanced stretches found in this chapter.

INTERMEDIATE LYING KNEE EXTENSOR STRETCH



Execution

1. Lie on the left side of the body.
2. Bend the right knee, and bring the right heel within 4 to 6 inches (10 to 15 cm) of the buttocks.
3. Grasp the right ankle tightly, and pull the leg back close to your buttocks. However, do not bring the heel of the right ankle all the way to the buttocks.
4. At the same time, push the hip forward.
5. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right vastus intermedius, right rectus femoris, right psoas major, middle and upper right sartorius

Less-stretched muscles: Right vastus medialis, left vastus lateralis, right tensor fasciae latae, right pectineus, right iliacus, anterior right gluteus medius, right tibialis anterior, right extensor digitorum longus, right extensor hallucis longus

Stretch Notes

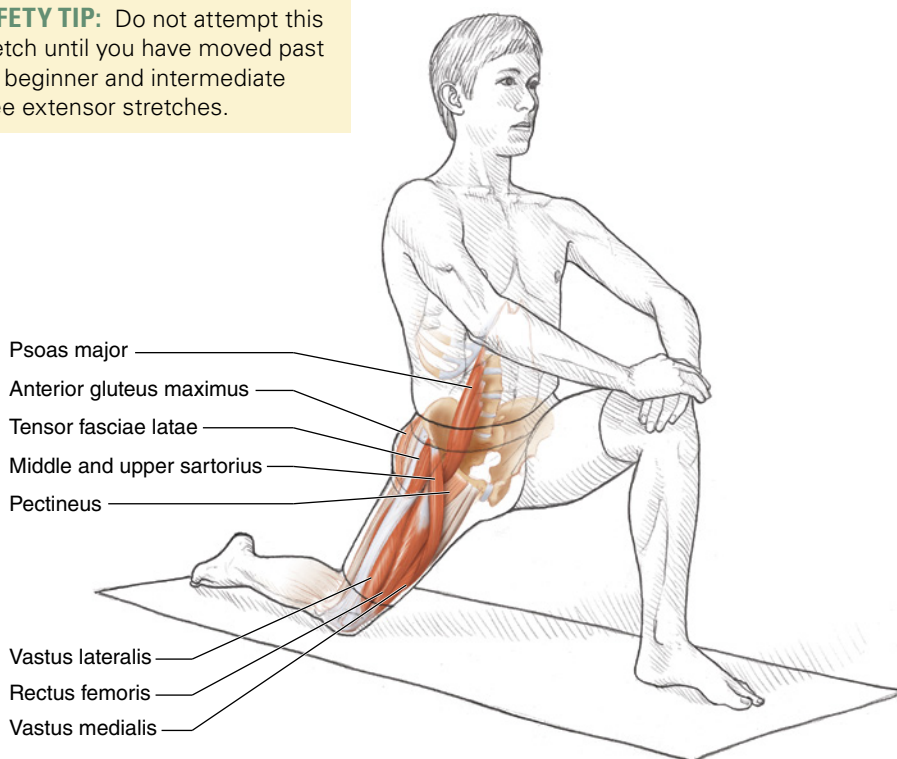
Injuries to the quadriceps muscles usually occur during an activity such as sprinting, jumping, or kicking, especially when the muscles are tight or unprepared for activity. This is yet another effective method of stretching the front thigh muscles. Although slightly more difficult than the beginner seated knee extensor stretch, this stretch still falls within the advanced beginner or intermediate category.

Because you perform this stretch while in a relaxed position, you have maximum control over the amount of stretch to the quadriceps muscles. In other words, this stretch allows you to concentrate solely on these thigh muscles while letting other muscles relax as much as possible.

Slowly pull the ankle in a more backward than upward direction while making sure the hips are also moving forward. Concentration should be greater on the forward hip movement than on the knee flexion (pulling the ankle toward the buttocks). As in any quadriceps muscle stretch, take extra care to prevent strain on the knee structure by overflexing the knee.

ADVANCED KNEELING KNEE EXTENSOR STRETCH

SAFETY TIP: Do not attempt this stretch until you have moved past the beginner and intermediate knee extensor stretches.



Execution

1. Step forward with the left leg, and bend the knee at about a 90-degree angle.
2. Keep the left knee positioned above the left ankle.
3. Extend the right leg behind the torso, and touch the floor with the right knee. The lower right leg lies on the floor.
4. Hold on to an object or place the hands on the left knee to maintain balance.
5. Move the hips forward, pushing the left knee in front of the left ankle and dorsiflexing that ankle.
6. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right quadriceps (vastus medialis, vastus intermedius, vastus lateralis, rectus femoris) middle and upper right sartorius, right psoas major, right iliacus, right tensor fasciae latae

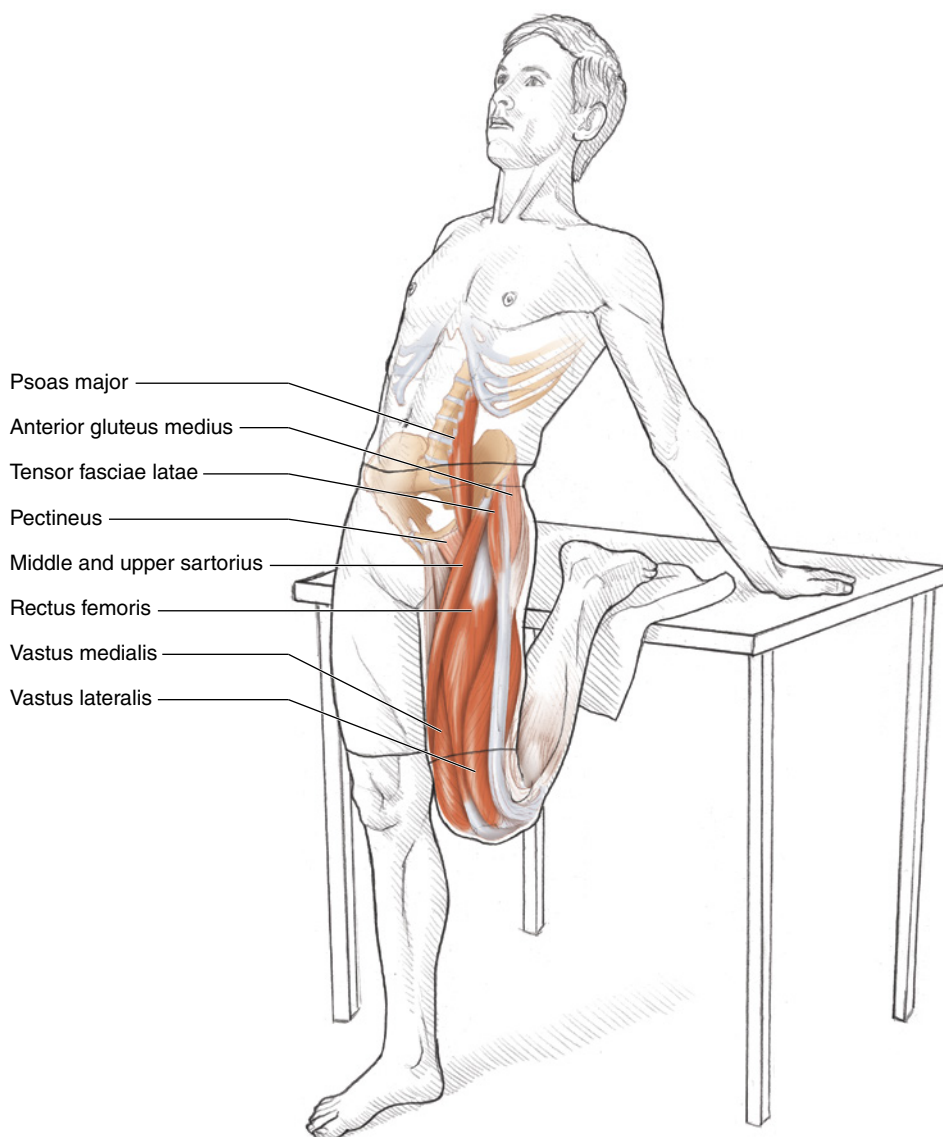
Less-stretched muscles: Right pectineus, anterior right gluteus maximus

Stretch Notes

The advanced kneeling knee extensor stretch is the quadriceps stretch most commonly used by athletes and nonathletes alike. Most people tend to have stronger but less flexible quadriceps muscles than hamstring muscles because of the tendency to stretch the hamstrings much more than the quadriceps. This creates an imbalance of strength and flexibility between the two muscle groups. To correct this imbalance, more emphasis needs to be placed on routinely stretching the quadriceps muscles.

When the right knee is extended behind the torso onto the floor, try to have a soft surface under the knee. This could be an exercise mat, grass, or even a pillow. This will minimize discomfort to the knee. When you move slowly into the stretched position, keep the left knee pointing forward. Do not let the left knee point to either side or let the right knee move along the surface of the floor. While the hips are placed in the forward direction, arching the back can increase the stretch on the muscles. This would stretch not only the quadriceps muscles but also the hip flexor muscles located in front of the pelvic area.

ADVANCED SUPPORTED STANDING KNEE EXTENSOR STRETCH



Execution

1. Stand with the back toward a padded table, bed, or soft platform that is below the height of the hips.
2. Balance your weight on the right leg, and bend the knee slightly.
3. Bend the left knee, and prop the left ankle on the rear support surface.
4. Place both hands on the rear support surface 6 to 12 inches (15 to 30 cm) behind the buttocks.
5. Move the torso back slowly so that the heel of the left foot comes as close to the buttocks as possible. Make sure the ankle and knee are comfortable.
6. Push the hips forward and simultaneously arch the back by bending the shoulders toward the buttocks.
7. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Left quadriceps (vastus medialis, vastus intermedius, vastus lateralis, rectus femoris) middle and upper right sartorius, left psoas major, left iliacus, left tensor fasciae latae

Less-stretched muscles: Left pectineus, anterior left gluteus medius

Stretch Notes

Knee stiffness can lead to injuries of the knee and of the quadriceps muscles and tendons. This is the most advanced stretch for the quadriceps muscles, and you must take extra care when attempting it. Because of the increased possibility of hyperflexing the knee, use this stretch only if you have very flexible muscles. By adhering to the following safety precautions, you can execute this stretch safely without injury.

While pulling the ankle slowly in a more backward than upward direction, concentrate on making sure your hips also move forward. This dual action stretches the hip flexor muscles located in front of the pelvic region as well as the quadriceps muscles. If you are experiencing soreness or tightness of either the lateral (outer) or medial (inner) side of the front thigh, consider placing most of the stretch emphasis on the medial muscles (vastus medialis and pectineus) by rotating the upper body away from the medial muscles (rotate the right side clockwise) when bending backward. To place most of the stretch emphasis on the lateral muscles (vastus lateralis and tensor fasciae latae), rotate the upper body away from the lateral muscles (rotate the right side counterclockwise) when bending backward.

For optimal results, it is important to brace both hands on the surface supporting the back. In addition, you should move your hips forward while

(continued)

Advanced Supported Standing Knee Extensor Stretch *(continued)*

carefully arching your back. This enables you to better control the amount of stretch being put on these muscles. Following these procedures maximizes stretch to the quadriceps muscles as well as to the hip flexor muscles located in front of the pelvic area. Yet another precaution for safety as well as for comfort is to pad the support on which the ankle is resting.

You might also consider moving the dorsal (top) part of the foot down to the edge of the padded support. This would bring additional benefits from the total stretch, because you also stretch the muscles in the anterior (front) part of the tibia bone in the lower leg. This is a powerful combination of multiple stretches.

In this stretch you are also able to change your trunk position, thus stretching the medial or lateral side of the thigh if you move your trunk in either a lateral (outer) or medial (inner) direction.